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CLEANING A STAND-ALONE ICE MAKING APPLIANCE

Abstract

A stand-alone ice making appliance includes an upper reservoir fluidly coupled to a lower reservoir via a first valve. An ice maker is disposed within the casing. A pump is disposed within the casing and is operable to flow water from the lower reservoir to upper reservoir and the ice maker. A controller is positioned in the casing. A method of operating the stand-alone ice making appliance includes opening the first valve between the upper reservoir and the lower reservoir. The method also includes draining the liquid from the upper reservoir to the lower reservoir and closing the first valve between the upper reservoir and the lower reservoir. The method further includes operating a descale or cleaning operation of the stand-alone ice making appliance.

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Background/Summary

FIELD OF THE INVENTION

[0001] The present subject matter relates generally to ice making appliances, and more particularly to systems and methods for cleaning components of stand-alone ice making appliances.

BACKGROUND OF THE INVENTION

[0002] Ice makers generally produce ice for use by consumers, such as in beverages, for cooling food items, etc. Certain refrigerator appliances include ice makers for producing ice. The ice maker can be positioned within the appliance's freezer chamber and direct ice into an ice bucket where the ice is stored within the freezer chamber. Such refrigerator appliances can also include a dispensing system for assisting a user with accessing ice produced by the refrigerator appliance's ice maker. However, the incorporation of ice makers into refrigerator appliances can have drawbacks, such as limits on the amount of ice that can be produced and the reliance on the refrigeration system of the refrigerator appliance to form the ice.

[0003] Stand-alone ice makers are separate from refrigerator appliances and provide independent ice supplies. Generally, liquid water is added to the stand-alone ice makers, and the ice makers operate to freeze the liquid water and form ice. Users frequently add tap water to the stand-alone ice makers. Tap water may include various impurities that negatively affect the appearance and/or taste of ice cubes formed from the tap water. Further, tap water may, over time, lead to scale buildup within the ice maker.

[0004] Traditional methods of cleaning some ice makers may include moving the whole ice machine closer to a sink or use a big bucket to drain the water used to clean and rinse, which is manually intensive. Accordingly, a system for cleaning an ice maker that removes the manual intensity of traditional methods would be advantageous.

BRIEF DESCRIPTION OF THE INVENTION

[0005] Aspects and advantages of the invention will be set forth in part in the following description, or may be obvious from the description, or may be learned through practice of the invention.

[0006] In one example embodiment, A stand-alone ice making appliance is provided. The stand-alone ice making appliance includes a casing and a user interface on the casing. A container is within the casing. A lower reservoir is removably mounted beneath the container. An upper reservoir is disposed within the casing. The upper reservoir is fluidly coupled to the lower reservoir via a first valve. An ice maker is disposed within the casing. A pump is disposed within the casing and is in fluid communication with each of the lower reservoir, the upper reservoir, and the ice maker. The pump is operable to flow water from the lower reservoir to the upper reservoir and the ice maker. A controller is positioned in the casing and is in signal communication with the first valve between the upper reservoir and the lower reservoir. The controller is configured to open the first valve between the upper reservoir and the lower reservoir. The controller is also configured to drain the liquid from the upper reservoir to the lower reservoir and close the first valve between the upper reservoir and the lower reservoir. The controller is further configured to operate a descale or cleaning operation of the stand-alone ice making appliance.

[0007] In another example embodiment, a method of operating a stand-alone ice making appliance is provided. The stand-alone ice making appliance includes a casing and a user interface on the casing. A container is within the casing. A lower reservoir is removably mounted beneath the container. An upper reservoir is disposed within the casing. The upper reservoir is fluidly coupled to the lower reservoir via a first valve. An ice maker is disposed within the casing. A pump is disposed within the casing and is in fluid communication with each of the lower reservoir, the upper reservoir, and the ice maker. The pump is operable to flow water from the lower reservoir to the upper reservoir and the ice maker. A controller is positioned in the casing and is in signal communication with the first valve between the upper reservoir and the lower reservoir. The method includes opening the first valve between the upper reservoir and the lower reservoir. The

method also includes draining the liquid from the upper reservoir to the lower reservoir and closing the first valve between the upper reservoir and the lower reservoir. The method further includes operating a descale operation of the stand-alone ice making appliance.

[0008] These and other features, aspects and advantages of the present invention will become better understood with reference to the following description and appended claims. The accompanying drawings, which are incorporated in and constitute a part of this specification, illustrate embodiments of the invention and, together with the description, serve to explain the principles of the invention.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0009] A full and enabling disclosure of the present invention, including the best mode thereof, directed to one of ordinary skill in the art, is set forth in the specification, which makes reference to the appended figures.

[0010] FIG. 1 is a perspective view of an appliance according to an example embodiment of the present disclosure.

[0011] FIG. 2 is a perspective section view of the example appliance of FIG. 1.

[0012] FIG. 3 is a rear perspective view of the example appliance of FIG. 1 with a casing of the example appliance removed to show interior components of the example appliance.

[0013] FIG. 4 is a schematic view of components of the example appliance of FIG. 1, according to aspects of the present disclosure.

[0014] FIG. 5 provides a flowchart of an example method of operating a cleaning vessel according to aspects of the present disclosure.

[0015] Repeat use of reference characters in the present specification and drawings is intended to represent the same or analogous features or elements of the present invention.

DETAILED DESCRIPTION

[0016] Reference now will be made in detail to embodiments of the invention, one or more examples of which are illustrated in the drawings. Each example is provided by way of explanation of the invention, not limitation of the invention. In fact, it will be apparent to those skilled in the art that various modifications and variations can be made in the present invention without departing from the scope or spirit of the invention. For instance, features illustrated or described as part of one embodiment can be used with another embodiment to yield a still further embodiment. Thus, it is intended that the present invention covers such modifications and variations as come within the scope of the appended claims and their equivalents.

[0017] As used herein, the terms “includes” and “including” are intended to be inclusive in a manner similar to the term “comprising.” Similarly, the term “or” is generally intended to be inclusive (i.e., “A or B” is intended to mean “A or B or both”). Approximating language, as used herein throughout the specification and claims, is applied to modify any quantitative representation that could permissibly vary without resulting in a change in the basic function to which it is related. Accordingly, a value modified by a term or terms, such as “about,” “approximately,” and “substantially,” are not to be limited to the precise value specified. In at least some instances, the approximating language may correspond to the precision of an instrument for measuring the value. For example, the approximating language may refer to being within a ten percent (10%) margin.

[0018] Referring now to FIGS. 1 through 3, one embodiment of an appliance **10** in accordance with the present disclosure is illustrated. As shown, appliance **10** is provided as a stand-alone ice making appliance embodiment. Appliance **10** includes an outer casing **12** which defines a primary opening **11** (e.g., first primary opening) and an internal cavity or volume **13**. Internal volume **13** generally at least partially houses various other components of the appliance therein **10**. Primary opening **11**

defined in outer casing **12** may extend internal volume **13** to an ambient environment. Through primary opening **11**, access (e.g., by a user) to the internal volume **13** may be permitted. Outer casing **12** further defines a vertical direction V, a lateral direction L, and a transverse direction T. The vertical direction V, lateral direction L, and transverse direction T are mutually perpendicular and form an orthogonal direction system.

[0019] A container **14** of appliance **10** is also illustrated. Container **14** defines a first storage volume **16** for the receipt and storage of ice **18** therein. A user of the appliance **10** may access ice **18** within the container **14** for consumption or other uses, as described in detail below. Container **14** may include multiple walls, including one or more sidewalls **20** and a base wall **22**, which may together define the first storage volume **16**. In example embodiments, at least one sidewall **20** may be formed in part from a clear, see-through (i.e., transparent, or translucent) material, such as a clear glass or plastic, such that a user can see into the first storage volume **16** and thus view ice **18** therein. For instance, at least one sidewall **20** may include a separate external panel and internal panel formed from a clear, see-through (i.e., transparent, or translucent) material, such as a clear glass or plastic. In some example embodiments, container **14** may include a handle **17**. In general, handle **17** may advantageously improve accessibility to ice **18** within container **14**. Further, in example embodiments, container **14** may be removable, such as from the outer casing **12**, by a user. This facilitates advantageous easy access by the user to ice within the container **14**, as discussed below. In general, a user interface **15** may be positioned on casing **12** above container **14**. The user interface **15** may generally include input selectors to be selected (e.g., by a user) for controlling the appliance **10**.

[0020] Appliances **10** in accordance with the present disclosure are advantageously stand-alone appliances, and thus are not connected to refrigerators or other appliances. Additionally, in example embodiments, such appliances are not connected to plumbing or another water source that is external to the appliance **10**, such as a refrigerator water source. Rather, in example embodiments, water is initially supplied to the appliance **10** manually by a user, such as by pouring water into reservoir **24** and/or an auxiliary reservoir **100**. Optionally, in example embodiments, reservoir **24** may be removable, such as from the outer casing **12**, by a user. This facilitates advantageous easy access by the user to reservoir **24** (e.g., in order to easily fill reservoir **24**), as discussed below.

[0021] Notably, appliances **10** as discussed herein include various features which allow the appliances **10** to be affordable and desirable to typical consumers. For example, the stand-alone feature reduces the cost associated with the appliance **10** and allows the consumer to position the appliance **10** at any suitable desired location, with the only requirement in some embodiments being access to an electrical source. In example embodiments, such as those shown in FIGS. **1** through **3**, the removable container **14** allows easy access to ice **18** within first storage volume **16** and allows the container **14** to be moved to a different position from the remainder of the appliance **10** for ice usage purposes.

[0022] As discussed herein, appliance **10** is configured to make nugget ice, which is becoming increasingly popular with consumers. Ice **18** may be nugget ice. Generally, nugget ice is ice that is maintained or stored (i.e., in first storage volume **16** of container **14**) at a temperature greater than the melting point of water or greater than about thirty-two degrees Fahrenheit. Accordingly, the ambient temperature of the environment surrounding the container **14** may be at a temperature greater than the melting point of water or greater than about thirty-two degrees Fahrenheit. In some embodiments, such temperature may be greater than forty degrees Fahrenheit, greater than fifty degrees Fahrenheit, or greater than sixty degrees Fahrenheit.

[0023] Still referring to FIGS. **1** through **3**, various components of appliance **10** in accordance with the present disclosure are illustrated. For example, as mentioned, appliance **10** includes a reservoir **24**, otherwise referred to as a lower reservoir **24**. The reservoir **24** defines a second storage volume **26** for the receipt and holding of water. Reservoir **24** may include multiple walls, including one or more sidewalls **28** and a base wall **30**, which may together define the second storage volume **26**. In

example embodiments, the reservoir **24** may be disposed below the container **14** along the vertical direction **V** defined for the appliance **10**, as shown. In the present example embodiment, reservoir **24** may be removable from appliance **10**, as will be further described hereinbelow.

[0024] As discussed, in example embodiments, water is provided to the reservoir **24** for use in forming ice. Accordingly, appliance **10** may further include a pump **32**. Pump **32** may be in fluid communication with the second storage volume **26**. For example, water may be flowable from the second storage volume **26** through a fluid outlet **31** defined in the reservoir **24**, such as in a sidewall **28** thereof, and may flow through a conduit to and through pump **32**. Pump **32** may, when activated, be operable to actively flow water from the second storage volume **26** therethrough and from the pump **32**.

[0025] Water actively flowed from the pump **32** may be flowed (e.g., through a suitable conduit) to a reservoir **34**, otherwise referred to as an upper reservoir **34**. For example, reservoir **34** may define a third storage volume **36**. In some embodiments, third storage volume **36** is defined by one or more sidewalls **38** and a base wall **40**. Third storage volume **36** may, for example, be in fluid communication with the pump **32** and may thus receive water that is actively flowed from the reservoir **24**, such as through the pump **32**. During operation, water may be flowed into the third storage volume **36** through an opening **44** defined in reservoir **34**.

[0026] Referring still to FIGS. **1** through **3**, reservoir **34** and third storage volume **36** thereof may receive and contain water to be provided to an ice maker **50** for the production of ice. Accordingly, third storage volume **36** may be in fluid communication with ice maker **50**. For example, water may be flowed, such as through an opening **42** and through suitable conduits, from third storage volume **36** to ice maker **50**.

[0027] Ice maker **50** generally receives water, such as from reservoir **34**, and freezes the water to form ice **18**. In example embodiments, ice maker **50** is a nugget ice maker, and in particular is an auger-style ice maker, although other suitable styles of ice makers and/or appliances are within the scope and spirit of the present disclosure. As shown, ice maker **50** may include a casing **52** into which water from third storage volume **36** is flowed. Casing **52** is thus in fluid communication with third storage volume **36**. For example, casing **52** may include one or more sidewalls **54** which may define an interior volume **56**, and an opening may be defined in a sidewall **54**. Water may be flowed from third storage volume **36** through the opening (such as via a suitable conduit) into the interior volume **56**.

[0028] As illustrated, an auger **60** may be disposed at least partially within the casing **52**. During operation, the auger **60** may rotate. Water within the casing **52** may at least partially freeze due to heat exchange, such as with a refrigeration system as discussed herein. The at least partially frozen water may be lifted by the auger **60** from casing **52**. Further, in example embodiments, the at least partially frozen water may be directed by auger **60** to and through an extruder **62**. The extruder **62** may extrude the at least partially frozen water to form ice, such as nuggets of ice **18**.

[0029] Formed ice **18** may be provided by the ice maker **50** to container **14**, and may be received in the first storage volume **16** thereof. For example, ice **18** formed by auger **60** and/or extruder **62** may be provided to the container **14**. In example embodiments, appliance **10** may include a chute **70** for directing ice **18** produced by the ice maker **50** towards the first storage volume **16**. For example, as shown, chute **70** is generally positioned above container **14** along the vertical direction **V**. Thus, ice can slide off of chute **70** and drop into storage volume **16** of container **14**. Chute **70** may, as shown, extend between ice maker **50** and container **14**, and may include a body **72**, which defines a passage **74** therethrough. Ice **18** may be directed from the ice maker **50** (such as from the auger **60** and/or extruder **62**) through the passage **74** to the container **14**. In some embodiments, for example, a sweep **64**, which may be connected to and rotate with the auger, may contact the ice emerging through the extruder **62** from the auger **60** and direct the ice **18** through the passage **74** to the container **14**.

[0030] As discussed, water within the casing **52** may at least partially freeze due to heat exchange,

such as with a refrigeration system. In example embodiments, ice maker **50** may include a sealed refrigeration system **80**. The sealed refrigeration system **80** may be in thermal communication with the casing **52** to remove heat from the casing **52** and interior volume **56** thereof, thus facilitating freezing of water therein to form ice. Sealed refrigeration system **80** may, for example, include a compressor **82**, a condenser **84**, a throttling device **86**, and an evaporator **88**. Evaporator **88** may, for example, be in thermal communication with the casing **52** in order to remove heat from the interior volume **56** and water therein during operation of sealed system **80**. For example, evaporator **88** may at least partially surround the casing **52**. In particular, evaporator **88** may be a conduit coiled around and in contact with casing **52**, such as the sidewall(s) **54** thereof.

[0031] It should additionally be noted that, in example embodiments, a controller **200** may be in operative communication with the sealed system **80**, such as with the compressor **82** thereof, and may activate the sealed system **80** as desired or required for ice making purposes.

[0032] For example, appliance **10** may include a controller **200** disposed within casing **312**. For example, controller **200** may be generally configured to facilitate operation of appliance **10**. In this regard, user interface **15** may be in communication with controller **200** such that controller **200** may receive control inputs from user interface **15** and may otherwise regulate operation of appliance **10**. For example, signals generated by controller **200** may operate appliance **10**, including any or all system components, subsystems, or interconnected devices, in response to the position of user interface **15** and other control commands. Specifically, draining and/or pumping fluid to/from appliance **10** may occur in response to a user input on user interface **15**. For example, the user interface **15** may receive the user input, e.g., a button press, a touch on a touchscreen interface, etc., and the user interface **15** may generate a corresponding signal in response to the user input and such signal may be transmitted to the controller **200**. Other components of appliance **10** may be in communication with controller **200** via, for example, one or more signal lines or shared communication busses. In this manner, Input/Output (“I/O”) signals may be routed between controller **200** and various operational components of appliance **10**.

[0033] As used herein, the terms “processing device,” “computing device,” “controller,” or the like may generally refer to any suitable processing device, such as a general or special purpose microprocessor, a microcontroller, an integrated circuit, an application specific integrated circuit (ASIC), a digital signal processor (DSP), a field-programmable gate array (FPGA), a logic device, one or more central processing units (CPUs), a graphics processing units (GPUs), processing units performing other specialized calculations, semiconductor devices, etc. In addition, these “controllers” are not necessarily restricted to a single element but may include any suitable number, type, and configuration of processing devices integrated in any suitable manner to facilitate appliance operation. Alternatively, controller **200** may be constructed without using a microprocessor, e.g., using a combination of discrete analog and/or digital logic circuitry (such as switches, amplifiers, integrators, comparators, flip-flops, AND/OR gates, and the like) to perform control functionality instead of relying upon software.

[0034] Controller **200** may include, or be associated with, one or more memory elements or non-transitory computer-readable storage mediums, such as RAM, ROM, EEPROM, EPROM, flash memory devices, magnetic disks, or other suitable memory devices (including combinations thereof). These memory devices may be a separate component from the processor or may be included onboard within the processor. In addition, these memory devices can store information and/or data accessible by the one or more processors, including instructions that can be executed by the one or more processors. It should be appreciated that the instructions can be software written in any suitable programming language or can be implemented in hardware. Additionally, or alternatively, the instructions can be executed logically and/or virtually using separate threads on one or more processors.

[0035] For example, controller **200** may be operable to execute programming instructions or micro-control code associated with an operating cycle of appliance **10**. In this regard, the instructions may

be software or any set of instructions that when executed by the processing device, cause the processing device to perform operations, such as running one or more software applications, displaying a user interface, receiving user input, processing user input, etc. Moreover, it should be noted that controller **200** as disclosed herein is capable of and may be operable to perform any methods, method steps, or portions of methods as disclosed herein. For example, in some embodiments, methods disclosed herein may be embodied in programming instructions stored in the memory and executed by controller **200**.

[0036] The memory devices may also store data that can be retrieved, manipulated, created, or stored by the one or more processors or portions of controller **200**. The data can include, for instance, data to facilitate performance of methods described herein. The data can be stored locally (e.g., on controller **200**) in one or more databases and/or may be split up so that the data is stored in multiple locations. In addition, or alternatively, the one or more database(s) can be connected to controller **200** through any suitable network(s), such as through a high bandwidth local area network (LAN) or wide area network (WAN). In this regard, for example, controller **200** may further include a communication module or interface that may be used to communicate with one or more other component(s) of appliance **10**, controller **200**, an external appliance controller, or any other suitable device, e.g., via any suitable communication lines or network(s) and using any suitable communication protocol. The communication interface can include any suitable components for interfacing with one or more network(s), including for example, transmitters, receivers, ports, controllers, antennas, or other suitable components.

[0037] In example embodiments, controller **200** is in operative communication with the pump **32**. Such operative communication may be via a wired or wireless connection, and may facilitate the transmittal and/or receipt of signals by the controller **200** and pump **32**. Controller **200** may be configured to activate the pump **32** to actively flow water. For example, controller **200** may activate the pump **32** to actively flow water therethrough when, for example, reservoir **34** requires water. A suitable sensor(s), for example, may be provided in the third storage volume **36**. The sensor(s) may be in operative communication with the controller **200** and may be configured to transmit signals to the controller **200**, which indicate whether or not additional water is desired in the reservoir **34**. When controller **200** receives a signal that water is desired, controller **200** may send a signal to pump **32** to activate pump **32**.

[0038] As shown in FIG. **1**, appliance **10** may also include an auxiliary water reservoir **100**. A height HWR of auxiliary water reservoir **100** may be about equal to a height HC of casing **12**. Thus, the appearance of auxiliary water reservoir **100** may complement casing **12**. Auxiliary water reservoir **100** may be disposed outside of casing **12**. For example, auxiliary water reservoir **100** may be mounted at the side of casing **12**. Thus, while most components of appliance **10** are housed within casing **12**, auxiliary water reservoir **100** is positioned outside of casing **12**. In certain example embodiments, auxiliary water reservoir **100** may include a base **110** and a container **120**. Base **110** may be attached to casing **12**, e.g., at the side of casing **12** adjacent to the bottom of casing **12**. For instance, base **110** may be clipped, fastened, etc. to casing **12**.

[0039] Auxiliary water reservoir **100** may be in fluid communication with a water tank within casing **12** such that water within auxiliary water reservoir **100** is flowable to the water tank. For example, a flexible tubing conduit, or a supply line may extend from auxiliary water reservoir **100** to reservoir **24**, whereby water from within auxiliary water reservoir **100** may flow from auxiliary water reservoir **100** into second storage volume **26**. It will be understood that appliance **10** may be plumbed in any other suitable manner to deliver water from auxiliary water reservoir **100** into casing **12** for use with ice maker **50** in alternative example embodiments.

[0040] Referring now to FIG. **4**, provided is a schematic including various components of appliance **10**. In particular, shown is upper reservoir **34** fluidly coupled to lower reservoir **24** via a first valve **306**. Furthermore, controller **200** may be in operative communication with first valve **306**. For example, controller **200** may generally be able to open and close first valve **306**, and drain

upper reservoir **34** into lower reservoir **24**, in response to user input(s) on user interface **15** (FIG. **1**). For example, when first valve **316** is open, fluid may drain via gravity from upper reservoir **34** to lower reservoir **24**. In some example embodiments, first valve **306** may include a siphon extending into upper reservoir **34**, thereby fluid may be siphoned from upper reservoir **34** into the lower reservoir **24**. During normal operation of appliance **10**, water hardness minerals from the fluid used to make ice may concentrate within ice maker **50** and upper reservoir **34**. Accordingly, controller **200** may generally be configured to operate a descale operation. For example, the descale operation may generally include pumping fluid, such as a descale solution (vinegar), from lower reservoir **24** to upper reservoir **34**, through ice maker **50**, back into upper reservoir **34**.

[0041] Moreover, appliance **10** may include a second valve **302** and a third valve **304**. In particular, second valve **302** may be positioned at a drain hole **19** (FIG. **2**) defined in container **14**, e.g., generally between container **14** and lower reservoir **24**, and third valve **304** may be positioned between lower reservoir **24** and pump **32**. For example, second valve **302** and third valve **304** may be check valves or poppet valves, e.g., second valve **302** and third valve **304** may be configured to close in response to a user removing lower reservoir **24**, reducing leakage of fluid from appliance **10** when lower reservoir **24** is removed.

[0042] Turning to FIG. **5**, a flowchart of an example method (e.g., method **500**) of operating appliance **10** will be described. Although the discussion below refers to the example method **500** of operating appliance **10**, one skilled in the art will appreciate that the example method **500** is applicable to the operation of a variety of other appliances, such as other possible variations of the appliance. In example embodiments, the various method steps as disclosed herein may be performed (e.g., in whole or part) by controller **200**, or another, separate, dedicated controller.

[0043] FIG. **5** depicts steps performed in a particular order for the purpose of illustration and discussion. Those of ordinary skill in the art, using the disclosures provided herein, will understand that (except as otherwise indicated) various example methods as may be disclosed herein are not mutually exclusive with each other, e.g., aspects of any one example method may be combined with aspects of any other example method, such that features illustrated or described as part of one embodiment can be used with another embodiment to yield a still further embodiment. Moreover, the steps of method **500** may be modified, adapted, rearranged, omitted, interchanged, or expanded in various ways without deviating from the scope of the present disclosure.

[0044] Referring now to FIG. **5**, at (**510**), method **500** may generally include opening first valve **306** between upper reservoir **34** and lower reservoir **24**. For example, first valve **306** may be opened via signal communication with controller **200** in response to receiving a user input at user interface **15**. At (**520**), method **500** may generally include draining fluid from upper reservoir **34** to lower reservoir **24**. For example, as a result of opening first valve **306**, fluid held within upper reservoir **34** may drain via gravity to lower reservoir **24**. In embodiments including the siphon, the method may include siphoning fluid from upper reservoir **34** into lower reservoir **24**. In other words, the siphon action may be controlled via activating pump **32** in response to upper reservoir **34** being full of fluid, thereby flushing fluid from upper reservoir **34** and flushing the concentrated water hardness minerals down to lower reservoir **24**. As stated above, during normal operation of appliance **10**, water hardness minerals separated from the fluid used to make ice may accumulate within ice maker **50** and upper reservoir **34**. While draining fluid from upper reservoir **34** to lower reservoir **24**, water hardness minerals and contaminants may be flushed out of upper reservoir **34**.

[0045] At (**530**), method **500** may generally include closing first valve **306** between upper reservoir **34** and lower reservoir **24**. With first valve **306** closed after the fluid and loose sediments have accumulated in lower reservoir **24**, a user may remove lower reservoir **24** and discard the contents of reservoir **24** in any suitable water receptacle, such as a kitchen sink, utility sink/tub, or waste water drain. With the contents discarded, the user may fill reservoir **24** with descaling solution, e.g., white vinegar and/or other similar solutions, such as any suitable acidic solution, including mixtures of one or more such solutions, which may, in some embodiments, be diluted with water,

and replace the reservoir **24** back into appliance **10**.

[0046] With reservoir **24** replaced in appliance **10**, at (**540**), method **500** may generally include operating the descale operation of appliance **10**. For example, as stated above, the descale operation may generally include pumping fluid, such as vinegar, from lower reservoir **24** to upper reservoir **34**, through ice maker **50**, back into upper reservoir **34**. After the descale operation, method **500** may generally include repeating steps **510**, **520**, and **530**, such that first valve **306** may be opened, the descale solution and additional sediments removed from ice maker **50** may drain from upper reservoir **34** into the lower reservoir **24**, and first valve **306** may be closed. Accordingly, after the descaling operation has been completed and the descaling solution and additional water hardness minerals have accumulated in lower reservoir **24**, the user may remove lower reservoir **24** and discard the contents of reservoir **24** in any suitable water receptacle.

[0047] Furthermore, after completing the descale operation and discarding the used descaling solution, the user may fill reservoir **24** with rinse fluid, such as water, and replace the reservoir **24** back into appliance **10**. At this point, the descale operation may be repeated with the rinse fluid so that the appliance is thoroughly rinsed before returning to normal operation. For example, the descale operation with the rinse water may be repeated between one and five times, such as between one and four times, such as between one and three times in order to thoroughly rinse appliance **10**. As such, after rinsing is complete, the user may fill reservoir **24** with tap water, or any suitable fluid for making nugget ice in the normal operation of appliance **10**, and replace reservoir **24** into appliance **10**.

[0048] Additionally or alternatively, some example embodiments may include opening first valve **306**, or the siphon (in certain example embodiments), when the fluid level in lower reservoir **24** is below a threshold, such as below thirty percent full, such as below twenty percent full, such as below ten percent full, in order to dump fluid from upper reservoir **34** into lower reservoir **24**. The user may then remove lower reservoir **24** and discard the fluid which includes the concentrated hardness minerals from upper reservoir **34**, and then the user may refill the tank. Discarding the fluid in lower reservoir **24** when the water level is below the threshold may reduce hardness minerals from accumulating to the point of scaling within ice maker **50**.

[0049] As may be seen from the above, provided may be a process of cleaning and descaling an ice making appliance by draining the upper reservoir, where hardness minerals may be concentrated, to a removable lower reservoir. A valve above the lower reservoir may be connected to a drain from the ice bin and another valve at the bottom of the lower reservoir may be connected to the pump. A drain line with an actuated valve may be positioned between the upper water reservoir and the lower reservoir and may be opened by a controller during the cleaning or descaling cycle. This process of cleaning an ice making appliance may advantageously reduce the manual intensity of traditional methods of cleaning/descaling ice making appliances.

[0050] This written description uses examples to disclose the invention, including the best mode, and also to enable any person skilled in the art to practice the invention, including making and using any devices or systems and performing any incorporated methods. The patentable scope of the invention is defined by the claims, and may include other examples that occur to those skilled in the art. Such other examples are intended to be within the scope of the claims if they include structural elements that do not differ from the literal language of the claims, or if they include equivalent structural elements with insubstantial differences from the literal languages of the claims.

Claims

1. A stand-alone ice making appliance, comprising: a casing; a user interface on the casing; a container within the casing; a lower reservoir removably mounted beneath the container; an upper reservoir disposed within the casing, the upper reservoir fluidly coupled to the lower reservoir via a

first valve; an ice maker disposed within the casing; a pump disposed within the casing, the pump in fluid communication with each of the lower reservoir, the upper reservoir and the ice maker, the pump operable to flow water from the lower reservoir to the upper reservoir and the ice maker; and a controller positioned in the casing, the controller in signal communication with the first valve between the upper reservoir and the lower reservoir, the controller configured to: open the first valve between the upper reservoir and the lower reservoir; drain fluid from the upper reservoir to the lower reservoir; close the first valve between the upper reservoir and the lower reservoir; and operate a descale or cleaning operation of the stand-alone ice making appliance.

2. The stand-alone ice making appliance of claim 1, wherein the first valve comprises a siphon extending into the upper reservoir, whereby fluid is siphoned from the upper reservoir into the lower reservoir.

3. The stand-alone ice making appliance of claim 1, wherein the container within the casing comprises a second valve fluidly coupled to a drain hole in the container.

4. The stand-alone ice making appliance of claim 3, wherein the second valve comprises a poppet valve configured to close in response to removal of the lower reservoir.

5. The stand-alone ice making appliance of claim 1, wherein the lower reservoir comprises a third valve fluidly coupled between the lower reservoir and the pump.

6. The stand-alone ice making appliance of claim 5, wherein the third valve comprises a poppet valve configured to close in response to removal of the lower reservoir.

7. The stand-alone ice making appliance of claim 1, wherein operating the descale operation further comprises the controller configured to pump fluid from the lower reservoir into the upper reservoir and through the ice maker.

8. The stand-alone ice making appliance of claim 7, further comprising, after the descale operation, the controller configured to: open the first valve between the upper reservoir and the lower reservoir; drain fluid from the upper reservoir to the lower reservoir; and close the first valve between the upper reservoir and the lower reservoir.

9. The stand-alone ice making appliance of claim 1, wherein the first valve is positioned between the upper reservoir and the lower reservoir such that fluid drains via gravity when the first valve is open.

10. A method of operating a stand-alone ice making appliance, the stand-alone ice making appliance comprising a casing, a user interface on the casing, a container within the casing, a lower reservoir removably mounted beneath the container, an upper reservoir disposed within the casing, the upper reservoir fluidly coupled to the lower reservoir via a first valve, an ice maker disposed within the casing, a pump disposed within the casing, the pump in fluid communication with each of the lower reservoir, the upper reservoir and the ice maker, the pump operable to flow water from the lower reservoir to the upper reservoir and the ice maker, and a controller positioned in the casing, the controller in signal communication with the first valve between the upper reservoir and the lower reservoir, the method comprising: opening the first valve between the upper reservoir and the lower reservoir; draining fluid from the upper reservoir to the lower reservoir; closing the first valve between the upper reservoir and the lower reservoir; and operating a descale operation of the stand-alone ice making appliance.

11. The method of claim 10, wherein the first valve comprises a siphon extending into the upper reservoir, whereby the method includes siphoning fluid from the upper reservoir into the lower reservoir.

12. The method of claim 10, wherein the container within the casing comprises a second valve fluidly coupled to a drain hole in the container.

13. The method of claim 12, wherein the second valve comprises a poppet valve configured to close in response to removal of the lower reservoir.

14. The method of claim 10, wherein the lower reservoir comprises a third valve fluidly coupled between the lower reservoir and the pump.

- 15.** The method of claim 14, wherein the third valve comprises a poppet valve configured to close in response to removal of the lower reservoir.
- 16.** The method of claim 10, further comprising pumping fluid from lower reservoir into the upper reservoir and through the ice maker during the descale operation.
- 17.** The method of claim 16, further comprising, after the descale operation: opening the first valve between the upper reservoir and the lower reservoir; draining fluid from the upper reservoir to the lower reservoir; and closing the first valve between the upper reservoir and the lower reservoir.
- 18.** The method of claim 10, wherein the first valve is positioned between the upper reservoir and the lower reservoir such that fluid drains via gravity when the first valve is open.
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